

Manipal Hackathon 2026



Team Name:	Vanishing Dose (Team Aman)
Track:	Healthcare (SDG 3: Good Health and Well-being)
Problem Statement:	P01: The Vanishing Dose - Detecting Medication Non-Adherence Without Asking
Solution Title:	Vanishing Dose: Multi-Signal Clinical Decision Support & Adherence Telemetry Engine
Team Members:	Aman, Tanish Kuvar
Working Prototype:	github.com/tanishkuvar11/Vanishing-Dose · Live Demo: localhost:5173/simulator
Evaluation Target:	Round 1 Ideation + Section 8.6 Prototype Bonus Submission

Solution



THE CLINICAL BLIND SPOT

The Lethal Reflex: Dose Escalation for Missed Pills

- **The Trap:** When clinic readings stay high, doctors reflexively double the dose. If the real cause was missed pills, escalating strength causes severe hypotension, bradycardia, or ICU admission when the patient resumes pills at home.
- **White-Coat Adherence:** Patients take pills frantically 2 to 3 days before visits. Readings look normal today, concealing weeks of unmedicated gaps.

Clinical Risk: Fatal Outpatient Toxicity

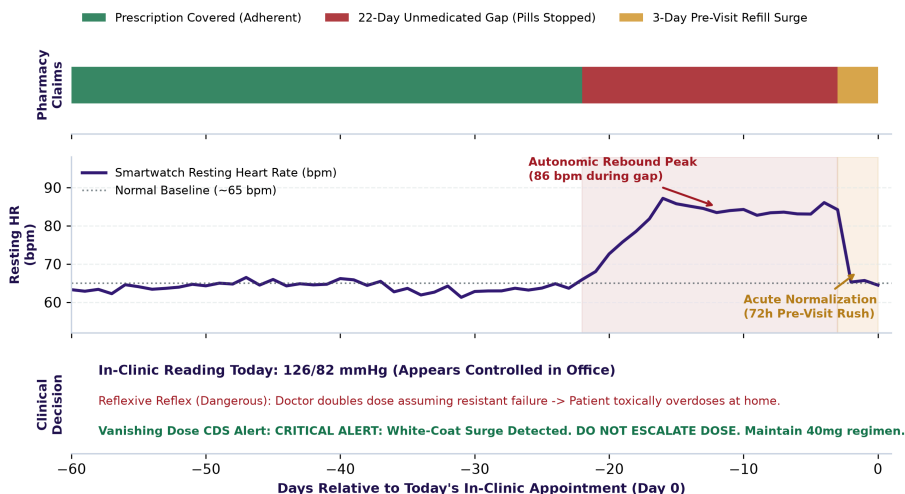
PROTOTYPE BONUS (PILLAR 6)

Live Working Software Prototype (Rulebook Section 6.4)

- **Interactive Sandbox:** 5 presets with 5 live sliders & real-time waveforms.
- **Doctor EHR Point-of-Care Screen:** Epic/Cerner simulated view with 1-click orders (Maintain Dose, Send Co-Pay Aid, Order Smart Cap, Auto-Insert Note).
- **Patient Mobile Companion:** CarePulse app with \$15 co-pay vouchers.

Public GitHub: [tanishkuvar11/Vanishing-Dose](https://github.com/tanishkuvar11/Vanishing-Dose)

Multi-Signal Telemetry: Exposing White-Coat Adherence & Preventing Toxic Dose Escalation



Technical Implementation



PIPELINE & PROVENANCE

CMS Claims Pipeline & Rigorous Censoring

- **8-Stage Python Pipeline:** 116,352 real Medicare beneficiaries from CMS DE-SynPUF files. NDCs standardized to active ingredients via RxNorm.
- **Hospital Stay Censoring:** Computes daily Proportion of Days Covered (PDC) while strictly censoring inpatient stays so institutional care is never mislabeled.

Real CMS Medicare Data Foundation

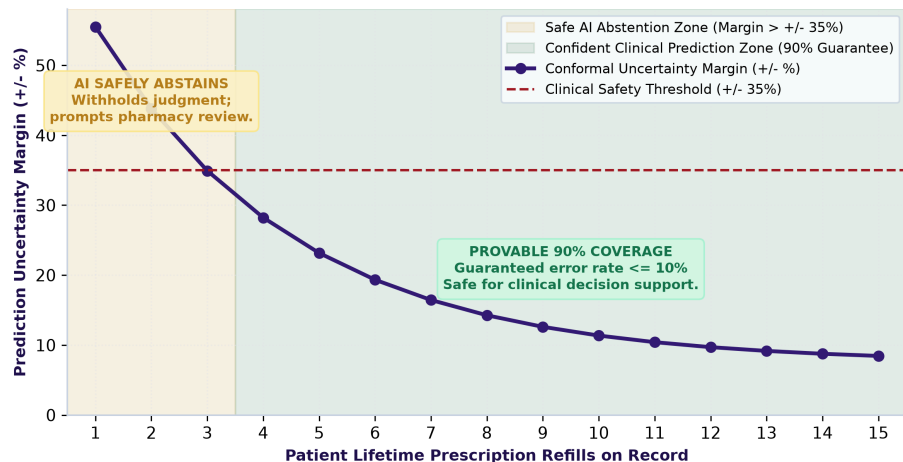
SAFE AI ABSTENTION

Split-Conformal Prediction: Zero Hallucination

- **Provable 90% Coverage:** Distribution-free calibration with finite-sample guarantees ($\alpha = 0.10$). Eliminates reckless guesses on sparse data.
- **Refusal to Predict:** For sparse records (<4 refills), uncertainty bounds exceed $\pm 35\%$. The AI safely withholds judgment and prompts doctor review.
- **EHR Standards:** Built around HL7 FHIR US Core and CDS Hooks.

HL7 FHIR & CDS Hooks Compliant

Split-Conformal Calibration: Eliminating AI Hallucination on Sparse Health Records



Feasibility



POINT-OF-CARE FIT

15-Second Doctor Workflow (Zero Extra Hardware)

- **Zero Dedicated Hardware:** Uses patient-owned consumer smartwatches (Apple HealthKit, Google Health Connect) and real-time pharmacy benefit claims.
- **15-Second Point-of-Care Actions:** Operates directly in the EHR consultation window. One click defers dangerous escalations and auto-populates notes.

Zero Clinic Friction / Instant Adoption

HEALTH EQUITY (SDG 3)

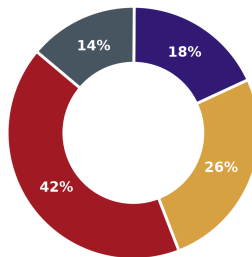
Non-Punitive Compassionate Support

- **Dismantling Patient Blame:** Over 40% of missed refills stem from out-of-pocket costs and transport deserts. We replace scolding with practical aid.
- **12-Month Execution Roadmap:** Q1: 10k patient retrospective IRB; Q2: 500-patient clinic pilot; Q3: CE/FDA SaMD & Epic App Orchard launch.

Empathetic Care / High Patient Retention

Health Equity & Feasibility: Replacing Patient Interrogation with Friction Relief

Why Patients Miss Doses
(Published Clinical Literature)



■ Prescription Cost & Co-pays
■ Routine Friction & Forgetting
■ Adverse Side Effects
■ Pharmacy Deserts & Transport

Vanishing Dose Targeted Health Equity Interventions

42% Cost Barrier -> \$15 Automated Co-Pay Discount Voucher

Reduces monthly out-of-pocket copay to under 10 dollars; auto-dispatched to pharmacy.

26% Habit Friction -> Bluetooth Smart Bottle Dispenser

Tracks cap openings without manual logging; sets gentle smartphone reminders.

18% Side Effects -> Confidential Care Pharmacist Chat

Interactive symptom triaging (dizziness, fatigue) with doctor schedule adjustments.

14% Transport Barrier -> Free 2-Day Discreet Home Delivery

Eliminates physical pharmacy travel; maintains 80%+ Medication Possession Ratio.

Business Strategy



GTM & OUTREACH

Dual-Stakeholder Hospital & Payer Go-To-Market

- **Target Buyers:** Hospital Health Systems and ACOs facing readmission penalties under CMS HRRP; Medicare Advantage plans targeting 5-Star ratings.
- **Outreach Channels:** Direct sales to CMOs; Epic App Orchard and Cerner App Gallery; peer-reviewed clinical publications at ACC and AHA conferences.

B2B Enterprise Distribution

MONETISATION MODEL

Enterprise SaaS with Value-Based Shared Savings

- **Tiered SaaS Subscription:** \$2.50 to \$4.00 Per Member Per Month (PMPM) based on hospital covered lives.
- **Shared Savings:** 15% fee on verified hospital readmission penalty cuts.
- **Unit Economics:** 82% gross margins; LTV:CAC > 5.2x; payback < 6 months.

High Recurring Enterprise ARR

Hospital Financial Impact: Cost per 1,000 Chronic Cardiovascular Patients

